

Scholarly Literature Map for §1.1: Economic Motivation and Forecasting Problem

Research scope and the strongest literature anchors

The literature supports a five-paragraph §1.1 built around **real-time information, macroeconomic instability and target heterogeneity, sequential data releases, data vintages and revisions, and model uncertainty/adaptation**. The strongest sources are not all saying the same thing, which is useful: they let the paper motivate the empirical design without prejudging its results.

The foundational real-time-nowcasting reference is Giannone, Reichlin and Small (2008). Their central insight is precisely the one needed for MacroPulse: economic decisions are made with incomplete data; macroeconomic series arrive on different dates; the resulting panel has a “jagged edge”; and a nowcast should be revised as new information becomes available. Their framework explicitly evaluates the marginal effect of intra-month data releases on current-quarter GDP nowcasts. ¹ Bańbura, Giannone, Modugno and Reichlin’s *Handbook of Economic Forecasting* chapter generalises this conception of nowcasting as monitoring many releases and revising assessments when new observations differ from expectations. ² Bok et al. (2018) shows that this large-data, real-time approach became an established operational framework for macroeconomic monitoring, including the New York Fed Staff Nowcast. ³

The second pillar is that **publication timing is itself economically important**. Bańbura and Rünstler (2011) find that surveys and financial variables can appear particularly valuable for GDP forecasting once their more timely publication is correctly accounted for; their broader conclusion is that publication lags should enter forecast evaluation rather than being treated as an incidental data-management detail. ⁴ Angelini et al. (2011) similarly study early GDP estimates using monthly series with heterogeneous publication lags and find value in timely survey and other “soft” information. ⁵

A particularly important discovery for positioning MacroPulse is Aastveit et al. (2014). Using **U.S. real-time data**, they update GDP density nowcasts after every new release and find that the ranking of three model classes changes during the quarter; their combined density forecast nevertheless performs consistently well and beats simple model selection in their application. ⁶ This paper is very close to the conceptual territory of a release-stage-dependent forecasting policy. It therefore belongs in §1.1 not only as support, but also as a constraint on what novelty MacroPulse can credibly claim.

The third pillar is **real-time vintage analysis**. Croushore and Stark (2001) established the modern real-time-data framework by organising macroeconomic observations as historical vintages and demonstrating that revisions can alter forecast analysis. ⁷ Croushore’s later review emphasises that revisions affect forecasting, structural modelling, monetary-policy analysis and current analysis. ⁸ The important methodological point for MacroPulse is straightforward: a database observed today is not necessarily the database that a forecaster would have possessed at the historical forecast origin. Croushore’s handbook treatment makes exactly this distinction between current databases and the information actually facing a real-time forecaster. ⁹

The revisions literature also requires nuance. Aruoba (2008), studying major U.S. macroeconomic variables, finds that revisions need not have zero means and can be predictable from information available when preliminary observations are published. ¹⁰ Yet Faust, Rogers and Wright (2005) find little predictability specifically for U.S. GDP revisions even though GDP revisions are sizeable, while predictability is stronger in several other G7 countries. ¹¹ Therefore §1.1 should **not** say that U.S. GDP revisions are systematically predictable. The safer and better-supported claim is that macroeconomic observations can change materially across vintages, that the properties of those revisions vary by series and setting, and that forecast evaluation can consequently depend on the vintage convention.

The fourth pillar is **forecast instability and model uncertainty**. Stock and Watson (1996), examining thousands of U.S. macroeconomic forecasting relations, document substantial instability in a significant fraction of univariate and bivariate models. Importantly, they also find that adaptive procedures do not automatically solve the problem. ¹² Rossi's modern review reaches an especially relevant conclusion for MacroPulse: instability is widespread, time variation in forecast performance need not correspond one-for-one with detectable parameter breaks, and local measures of predictive performance can be more informative than averages over long evaluation periods. ¹³

Finally, the forecast-combination literature strongly argues that the paper needs combination benchmarks. Stock and Watson (2004) find that individual predictor forecasts are unstable over time and countries, while relatively simple combinations often improve on autoregressive benchmarks. ¹⁴ Timmermann's handbook survey reports the broader empirical regularity that combinations often outperform ex-ante model selection and that simple combinations can outperform theoretically elaborate weighting schemes because estimation error, misspecification and instability matter. ¹⁵ Dynamic-model-averaging research goes still further: Koop and Korobilis allow the entire inflation forecasting model to change over time, while Onorante and Raftery dynamically alter the active model space in a GDP-nowcasting application. ¹⁶

What this literature means for the MacroPulse research question

The literature supports writing the forecasting problem around an **information set indexed by forecast origin and information stage**, rather than treating the data available for a target period as fixed:

$$\mathcal{I}_{t,s},$$

where t denotes the target/forecast origin convention and s denotes a release stage. This is a natural formalisation of the staggered-release and jagged-edge problem studied by Giannone et al. and the publication-lag problem emphasised by Bańbura and Rünstler. ¹⁷

A candidate model m can then produce

$$\hat{y}_{t,s}^{(m)} = f_m(\mathcal{I}_{t,s}),$$

and the empirical question is not simply whether m_1 beats m_2 in an unconditional average. The literature gives strong reasons to ask whether relative performance varies with s , with the target being predicted, and through time. Stock and Watson document structural instability; Rossi stresses local forecasting performance; and Aastveit et al. directly demonstrate within-quarter changes in model rankings as releases accumulate. ¹⁸

This is also where the MacroPulse descriptive evidence becomes useful. The current-vintage exercise shows large differences in the distribution of the paper’s targets before and after 2020: for example, annualised monthly core-CPI inflation has a standard deviation of about **0.74 percentage points during 2015–2019 versus 2.50 thereafter**, while monthly payroll-change volatility rises from roughly **83 thousand to 2.44 million**, heavily influenced by the extraordinary pandemic labour-market observations. Real-GDP growth since 2015 includes approximately -32.8% annualised growth in 2020Q2 and $+29.9\%$ in 2020Q3. These are paper-specific descriptive facts, not evidence that any particular forecasting policy must win. They provide a good empirical bridge to the literature on instability and pandemic-era forecasting challenges, where nonlinear or more flexible models have been motivated by precisely such extreme observations. ¹⁹

A second distinction should be made between the **release-stage problem** and the **vintage problem**. The former asks which observations have arrived by stage s ; the latter asks what numerical values those observations carried at the historical forecast date. Conceptually,

$$\mathcal{I}_{t,s}^{RT} / \mathcal{I}_{t,s}^{LV} =$$

because later revisions can alter the historical observations used by a latest-vintage reconstruction. That distinction follows directly from the real-time-data literature. ²⁰

Your ALFRED validation gives a concrete paper-specific illustration. Across 2010Q1–2018Q3, the absolute difference between the January-2019 and January-2020 vintage measures of annualised quarterly GDP growth averages about **0.219 percentage points**, with a maximum of about **1.219 points**; the next pair of annual snapshots gives an average absolute difference of about **0.207 points**. The largest first comparison changes 2017Q4 growth from approximately 2.27% to 3.48% . Those numbers should be described only as **illustrative changes between selected ALFRED snapshots**. They are not estimates of the unconditional revision distribution, and they are not first-release-to-final revisions.

Exact citation-to-paragraph map

I recommend preserving the five-paragraph architecture. The map below is sufficiently precise that the eventual prose can be drafted directly from it without another literature search.

Paragraph	Function in §1.1	Claim sequence	Exact citations to attach
Opening paragraph	Define the economic forecasting problem as real-time information processing	Sentence 1: policymakers/ forecasters must assess current conditions before all relevant official observations are available. Sentence 2: macro series arrive asynchronously, producing an unbalanced or jagged-edge information set. Sentence 3: nowcasts therefore evolve as releases arrive. Sentence 4: introduce $\mathcal{I}_{t,s}$ and define s as an information/ release stage, not merely a conventional forecast horizon.	<code>\citep{giannone2008nowcasting,banbura2013realtimeflo</code> edge; Bańbura et al. support the broader real-time-flow interpretati

Paragraph	Function in §1.1	Claim sequence	Exact citations to attach
Target and regime paragraph	Explain why a single globally stable forecasting relationship is demanding	Sentence 1: macroeconomic predictive relations exhibit substantial instability. Sentence 2: instability concerns predictive performance, not merely detectable coefficient breaks. Sentence 3: insert one compact MacroPulse empirical example showing the exceptional volatility shift around 2020 across GDP/inflation/labour. Sentence 4: conclude that model performance should be treated as an empirical object allowed to vary across targets and time.	<pre>\cite{stockwatson1996instability,rossi2021instability} \filecite{turn0file1}</pre>

Paragraph	Function in §1.1	Claim sequence	Exact citations to attach
Release-stage paragraph	Establish why forecast performance may change as the information set fills in	Sentence 1: new macro releases have measurable marginal informational content. Sentence 2: publication lags alter the effective forecasting usefulness of predictors; timely soft/ financial information can matter early even when harder activity data eventually dominate. Sentence 3: model rankings have been shown to change within the quarter as information accumulates. Sentence 4: this motivates comparing fixed versus stage-conditioned forecasting policies under identical information sets.	<code>\citep{giannone2008nowcasting,banbura2011blackbox,aa</code>

Paragraph	Function in §1.1	Claim sequence	Exact citations to attach
Vintage paragraph	Establish why historically feasible data rather than today's revised database are required	Sentence 1: latest-vintage observations are not generally identical to those available to historical forecasters. Sentence 2: revision processes can be economically important and their statistical properties differ by variable. Sentence 3: use the ALFRED GDP example as an illustration. Sentence 4: define $\mathcal{I}_{t,s}^{RT}$ versus $\mathcal{I}_{t,s}^{LV}$ and say the formal empirical design will hold forecast origins/model rules fixed while varying the vintage convention.	<code>\cite{croushorestark2001realtime,croushore2011front</code> <code>faustetal2005newsnoise</code> if discussing predictability. ²⁴ Paper

		Sentence 1: unstable relative model performance creates model uncertainty. Sentence 2: combinations are an important benchmark because simple pooling has often been robust when individual forecasts are unstable. Sentence 3: acknowledge explicitly that prior U.S. real- time nowcasting work already finds within- quarter changes in model rankings and good performance from combination forecasts. Sentence 4: state the narrower MacroPulse contribution as a systematic common- design comparison across the paper's multiple targets, information stages and vintage conventions; do	
Research- design/ contribution paragraph	Move from motivation to what MacroPulse will test		\citep{stockwatson2004combination,timmermann2006comb 25

Paragraph	Function in §1.1	Claim sequence	Exact citations to attach
		not claim that stage dependence or adaptive model weighting itself is new.	

A good word allocation for the eventual section would be approximately **150–170 words per paragraph**, giving roughly 750–850 words before any equations or footnotes. That is enough to establish the forecasting problem without allowing §1.1 to become a miniature literature review.

The most important positioning result from the search

The literature search changes how I would frame the paper's contribution.

A claim such as:

“Existing nowcasting methods use a fixed model throughout the quarter, whereas MacroPulse allows the preferred model to vary with the release stage.”

would be unsafe. Aastveit et al. explicitly find that the ranking of model classes changes within the quarter, update forecasts after individual releases, and compare model selection with combination strategies. ²⁶ Mazzi et al. likewise find that the relative usefulness of hard and soft indicators changes as within-quarter information accumulates. ²⁷

Likewise, a generic claim that MacroPulse is novel because it “allows the model to change over time” would be too strong. Dynamic model averaging has long allowed entire forecasting specifications to change through time, including U.S. inflation forecasting, and Onorante and Raftery apply dynamic model averaging to a GDP-nowcasting problem. ¹⁶

Nor should the paper claim novelty simply because it uses **real-time vintages**. There is a mature literature on real-time macroeconomic datasets and real-time forecast evaluation, beginning well before the current project. ²⁸

The safer, potentially stronger positioning is the **joint experimental design**. Subject to the exact models and hypotheses in later sections, MacroPulse can be presented as asking whether a forecasting policy should depend jointly on

(target, information stage, historical performance)

when all competing methods are evaluated on a common pseudo-real-time release calendar, and separately measuring what changes when

$$\mathcal{I}_{t,s}^{RT}$$

is replaced by

$$\mathcal{I}_{t,s}^{LV}.$$

The literature clearly contains each component separately—release-by-release nowcasting, changing model rankings, model averaging, structural instability and vintage analysis—but the scholarly search conducted here does **not** establish that their particular combination in MacroPulse's multi-target architecture is unprecedented. The manuscript should therefore use wording such as **“we provide a unified evaluation”, “we jointly examine”, or “we evaluate within a common real-time design”**, rather than “for the first time”, unless a dedicated novelty review later establishes priority. The closest paper that reviewers are likely to invoke is Aastveit et al. (2014), so §1.1 should pre-emptively acknowledge it. ²⁶

There is also a substantive implication for the empirical benchmark set. Because Aastveit et al. find density combination superior to simple model selection in their U.S. GDP application, and Stock–Watson/Timmermann document broader robustness of forecast combinations, a MacroPulse stage-selection policy should be compared against at least one transparent combination benchmark rather than only individual models. ²⁹ Otherwise a reviewer could reasonably ask whether apparent stage-selection gains survive comparison with diversification across models.

Claim boundaries and wording that will survive peer review

Several distinctions from the literature should be carried directly into the prose.

“Information arrives sequentially” is strongly established. It is safe to say that macroeconomic nowcasting operates with staggered releases, mixed frequencies and an incomplete recent edge of the dataset. ²¹

“The informational value of predictors is stage-dependent” is also defensible, provided this means their usefulness can depend on publication timing and the information already observed. Bańbura and Rünstler show that publication lags materially affect measured forecasting usefulness, while Aastveit et al. show within-quarter model rankings changing as releases accumulate. ³⁰

“A stage-adaptive forecasting policy will outperform a fixed policy” is not established. It should remain an empirical hypothesis. Stock and Watson's instability evidence does not imply that adaptive techniques automatically improve forecasts, and Aastveit et al. actually find that combination can outperform simple selection. ³¹

“Latest-vintage backtests use information unavailable in real time” requires careful phrasing. Latest-vintage datasets can contain revisions not available at historical origins, so they do not replicate the historical information set. That is the defensible statement. Whether using them materially changes a specific forecast ranking is an empirical question. ²⁰

“U.S. GDP revisions are predictable” should not appear as a blanket statement. Aruoba finds predictable revisions among major U.S. macro series, but Faust et al. report relatively little predictability for U.S. GDP revisions specifically. ³² The paper only needs the weaker proposition that historical estimates can change across vintages and therefore must be reconstructed when the aim is pseudo-real-time evaluation.

The COVID evidence should motivate instability, not establish it by itself. Your data display a sharp post-2020 increase in volatility and huge outliers across several targets. ^{filecite turn0file1} The long-standing Stock–Watson and Rossi literatures establish the more general issue of unstable forecasting relations, while pandemic-era research shows why extreme observations created particular modelling difficulties. ³³

The ALFRED exercise is motivation, not the H3 test. The annual snapshots demonstrate that historical GDP-growth measurements can change materially. `\filecite{turn0file0}` A proper real-time-versus-latest-vintage counterfactual must instead hold forecast origin, target date and forecasting rule constant, changing only the admissible data vintage. That design follows naturally from the real-time-data literature. ²⁰

Reference set to place in `references.bib`

These are the references I would treat as the **core §1.1 bibliography**. Suggested BibTeX keys are included so the eventual prose can use stable `\cite{}` calls.

Suggested key	Reference and role
<code>giannone2008nowcasting</code>	Giannone, D., Reichlin, L. & Small, D. (2008), "Nowcasting: The real-time informational content of macroeconomic data", <i>Journal of Monetary Economics</i> , 55(4), 665–676. DOI 10.1016/j.jmoneco.2008.05.010. Primary source for staggered releases, jagged edge and marginal news content. ³⁴
<code>banbura2013realtimeflow</code>	Bañbura, M., Giannone, D., Modugno, M. & Reichlin, L. (2013), "Now-Casting and the Real-Time Data Flow", <i>Handbook of Economic Forecasting</i> , Vol. 2, 195–237. DOI 10.1016/B978-0-444-53683-9.00004-9. Canonical synthesis of real-time nowcasting. ³⁵
<code>banbura2011blackbox</code>	Bañbura, M. & Rünstler, G. (2011), "A look into the factor model black box: Publication lags and the role of hard and soft data in forecasting GDP", <i>International Journal of Forecasting</i> , 27(2), 333–346. Primary source for the importance of publication timing. ³⁶
<code>aastveit2014density</code>	Aastveit, K. A., Gerdrup, K. R., Jore, A. S. & Thorsrud, L. A. (2014), "Nowcasting GDP in Real Time: A Density Combination Approach", <i>Journal of Business & Economic Statistics</i> , 32(1), 48–68. DOI 10.1080/07350015.2013.844155. Closest prior art for changing model rankings during the release cycle and combination versus selection. ²⁶
<code>stockwatson1996instability</code>	Stock, J. H. & Watson, M. W. (1996), "Evidence on Structural Instability in Macroeconomic Time Series Relations", <i>Journal of Business & Economic Statistics</i> , 14(1), 11–30. DOI 10.1080/07350015.1996.10524626. Foundational empirical evidence on forecasting instability. ³⁷
<code>rossi2021instability</code>	Rossi, B. (2021), "Forecasting in the Presence of Instabilities: How We Know Whether Models Predict Well and How to Improve Them", <i>Journal of Economic Literature</i> , 59(4), 1135–1190. DOI 10.1257/jel.20201479. Modern authority on unstable predictive performance and local evaluation. ¹³

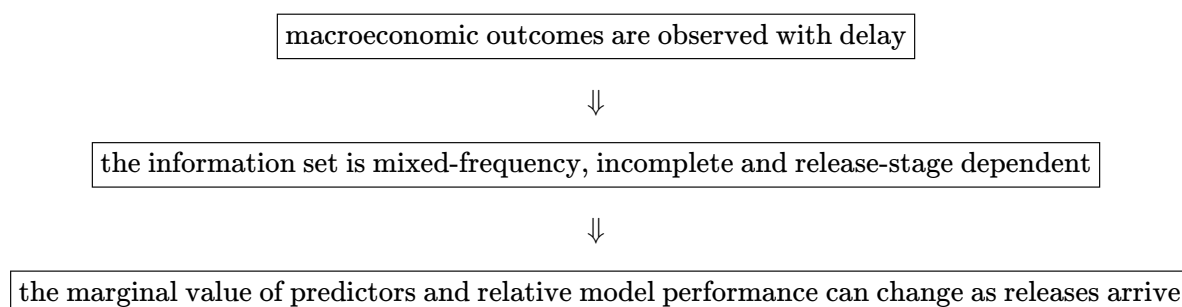
Suggested key	Reference and role
<code>croushorestark2001realtime</code>	Croushore, D. & Stark, T. (2001), "A Real-Time Data Set for Macroeconomists", <i>Journal of Econometrics</i> , 105(1), 111–130. DOI 10.1016/S0304-4076(01)00072-0. Foundational vintage-data citation. ³⁸
<code>croushore2011frontiers</code>	Croushore, D. (2011), "Frontiers of Real-Time Data Analysis", <i>Journal of Economic Literature</i> , 49(1), 72–100. DOI 10.1257/jel.49.1.72. Review citation for consequences of revisions across forecasting and policy analysis. ⁸
<code>aruoba2008revisions</code>	Aruoba, S. B. (2008), "Data Revisions Are Not Well Behaved", <i>Journal of Money, Credit and Banking</i> , 40(2–3), 319–340. DOI 10.1111/j.1538-4616.2008.00115.x. Evidence that revision properties need not be benign. ³⁹
<code>faustetal2005newsnoise</code>	Faust, J., Rogers, J. H. & Wright, J. H. (2005), "News and Noise in G-7 GDP Announcements", <i>Journal of Money, Credit and Banking</i> , 37(3), 403–419. Useful qualification: U.S. GDP revision predictability is comparatively limited. ⁴⁰
<code>stockwatson2004combination</code>	Stock, J. H. & Watson, M. W. (2004), "Combination forecasts of output growth in a seven-country data set", <i>Journal of Forecasting</i> , 23(6), 405–430. DOI 10.1002/for.928. Core combination/instability benchmark. ⁴¹
<code>timmermann2006combinations</code>	Timmermann, A. (2006), "Forecast Combinations", <i>Handbook of Economic Forecasting</i> , Vol. 1, 135–196. Canonical synthesis explaining why simple pooling can be robust to model uncertainty and instability. ¹⁵

Three **secondary/additional** references should be retained for later sections but need not all appear in §1.1: Angelini et al. (2011) for factor/bridge nowcasting with publication lags, Koop and Korobilis (2012) for dynamic model averaging in inflation forecasting, and Onorante and Raftery (2016) for dynamically changing model spaces in GDP nowcasting. ⁴²

The 2018 Bok et al. review is also useful when the manuscript later needs an institutional/modern-big-data bridge from the classical nowcasting literature to operational central-bank systems. ⁴³

Drafting blueprint to freeze before writing the LaTeX

The scholarly evidence now supports the following logical progression:



⇓

predictive relationships are also unstable across time and targets

⇓

forecast selection/combination should therefore be evaluated empirically rather than assumed fixed

while, independently,

historical macroeconomic observations are revised

implies

$$\mathcal{I}_{t,s}^{RT} / = \mathcal{I}_{t,s}^{LV}$$

and hence motivates a separate vintage counterfactual. The first chain is supported by the real-time-flow, publication-lag, instability and forecast-combination literatures; the second is supported by Croushore–Stark and the broader revision literature. ⁴⁴

The final paragraph of §1.1 should therefore **not** announce a presumed winning model. Its role should be to pose the research problem: given a common historically feasible information set, does the relative value of competing forecasting approaches vary systematically across targets and release stages; can an adaptive or combined policy exploit that variation out of sample; and how much of the measured forecasting performance changes when historically feasible vintages are replaced by latest-revised data? That formulation is consistent with what the literature establishes while leaving the answers to the empirical analysis. ⁴⁵

The literature search is therefore sufficiently mature to **freeze the five-paragraph source architecture before drafting**. The key editorial discipline for the later LaTeX version is to distinguish three things throughout: **established facts from the literature**, **descriptive facts generated by MacroPulse**, and **hypotheses that the paper will test**. The uploaded Stata results supply the second category—both the target-volatility evidence and the illustrative GDP-vintage differences—while the sources above supply the first. ⁴⁶

¹ ¹⁷ ²¹ ²³ ³⁴ ⁴⁴ Nowcasting: The real-time informational content of macroeconomic data - ScienceDirect

https://www.sciencedirect.com/science/article/pii/S0304393208000652?utm_source=chatgpt.com

² ³⁵ EconPapers: Now-Casting and the Real-Time Data Flow

https://econpapers.repec.org/RePEc%3Aaeee%3Aecofch%3A2-195?utm_source=chatgpt.com

³ ⁴³ Macroeconomic Nowcasting and Forecasting with Big Data | Annual Reviews

https://www.annualreviews.org/content/journals/10.1146/annurev-economics-080217-053214?utm_source=chatgpt.com

⁴ ³⁰ ³⁶ A look into the factor model black box: Publication lags and the role of hard and soft data in forecasting GDP

https://ideas.repec.org/a/eee/intfor/v27yi2p333-346.html?utm_source=chatgpt.com

⁵ ⁴² Short-term forecasts of euro area GDP growth

https://ideas.repec.org/a/wly/emjrn/v14y2011ipc25-c44.html?utm_source=chatgpt.com

- 6 26 29 45 **Nowcasting GDP in Real Time: A Density Combination Approach: Journal of Business & Economic Statistics: Vol 32, No 1**
https://www.tandfonline.com/doi/abs/10.1080/07350015.2013.844155?utm_source=chatgpt.com
- 7 20 24 28 38 **A real-time data set for macroeconomists - ScienceDirect**
https://www.sciencedirect.com/science/article/abs/pii/S0304407601000720?utm_source=chatgpt.com
- 8 **EconPapers: Frontiers of Real-Time Data Analysis**
https://econpapers.repec.org/RePEc%3Aaea%3Ajeclit%3Av%3A49%3Ay%3A2011%3Ai%3A1%3Ap%3A72-100?utm_source=chatgpt.com
- 9 **Forecasting with Real-Time Data Vintages | The Oxford Handbook of Economic Forecasting | Oxford Academic**
https://academic.oup.com/edited-volume/28323/chapter/215065822?utm_source=chatgpt.com
- 10 32 39 **Data Revisions Are Not Well Behaved - ARUOBA - 2008 - Journal of Money, Credit and Banking - Wiley Online Library**
https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1538-4616.2008.00115.x?utm_source=chatgpt.com
- 11 40 **EconPapers: News and Noise in G-7 GDP Announcements**
https://econpapers.repec.org/article/mcbjmoncb/v_3a37_3ay_3a2005_3ai_3a3_3ap_3a403-19.htm?utm_source=chatgpt.com
- 12 18 22 33 37 **Evidence on Structural Instability in Macroeconomic Time Series Relations: Journal of Business & Economic Statistics: Vol 14, No 1**
https://www.tandfonline.com/doi/abs/10.1080/07350015.1996.10524626?utm_source=chatgpt.com
- 13 **Forecasting in the Presence of Instabilities: How We Know Whether Models Predict Well and How to Improve Them - American Economic Association**
<https://www.aeaweb.org/articles?id=10.1257%2Fjel.20201479>
- 14 25 41 **Combination forecasts of output growth in a seven-country data set - Stock - 2004 - Journal of Forecasting - Wiley Online Library**
https://onlinelibrary.wiley.com/doi/abs/10.1002/for.928?utm_source=chatgpt.com
- 15 **EconPapers: Forecast Combinations**
https://econpapers.repec.org/RePEc%3Aeee%3Aecofch%3A1-04?utm_source=chatgpt.com
- 16 **FORECASTING INFLATION USING DYNAMIC MODEL AVERAGING* - Koop - 2012 - International Economic Review - Wiley Online Library**
https://onlinelibrary.wiley.com/doi/full/10.1111/j.1468-2354.2012.00704.x?utm_source=chatgpt.com
- 19 **Nowcasting in a pandemic using non-parametric mixed frequency VARs - ScienceDirect**
https://www.sciencedirect.com/science/article/pii/S0304407620303936?utm_source=chatgpt.com
- 27 **Density Nowcasts and Model Combination: Nowcasting Euro-Area GDP Growth over the 2008–09 Recession* - Mazzi - 2014 - Oxford Bulletin of Economics and Statistics - Wiley Online Library**
https://onlinelibrary.wiley.com/doi/10.1111/obes.12015?utm_source=chatgpt.com
- 31 **Evidence on Structural Instability in Macroeconomic Time Series Relations | NBER**
https://www.nber.org/papers/t0164?utm_source=chatgpt.com